

CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY



Lecture No -13



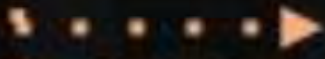
By- SATISH YADAV SIR

A yellow diamond-shaped sign with a black border and the text 'TOPICS TO BE COVERED' in bold black capital letters.

TOPICS TO BE COVERED

A horizontal dotted line with an arrow pointing to the right.

01 Basics of relations

A horizontal dotted line with an arrow pointing to the right.

02 Types of relations

A horizontal dotted line with an arrow pointing to the right.A vertical dotted line with an arrow pointing downwards.

03 Number Of relations

Relations

$[L, \vee, \wedge]$

$$1) a \vee a = a \quad a \wedge a = a.$$

$$2) a \vee b = b \vee a \quad a \wedge b = b \wedge a$$

$$3) a \vee (b \vee c) = (a \vee b) \vee c$$

$$a \wedge (b \wedge c) = (a \wedge b) \wedge c.$$

$$4) a \vee (a \wedge b) = a$$

$$a \wedge (a \vee b) = a$$

$$\underline{a} \wedge (\underline{a} \vee \text{---}) = a$$

$$a \vee (a \wedge \text{---}) = a$$

Relations



Bounded lattice:

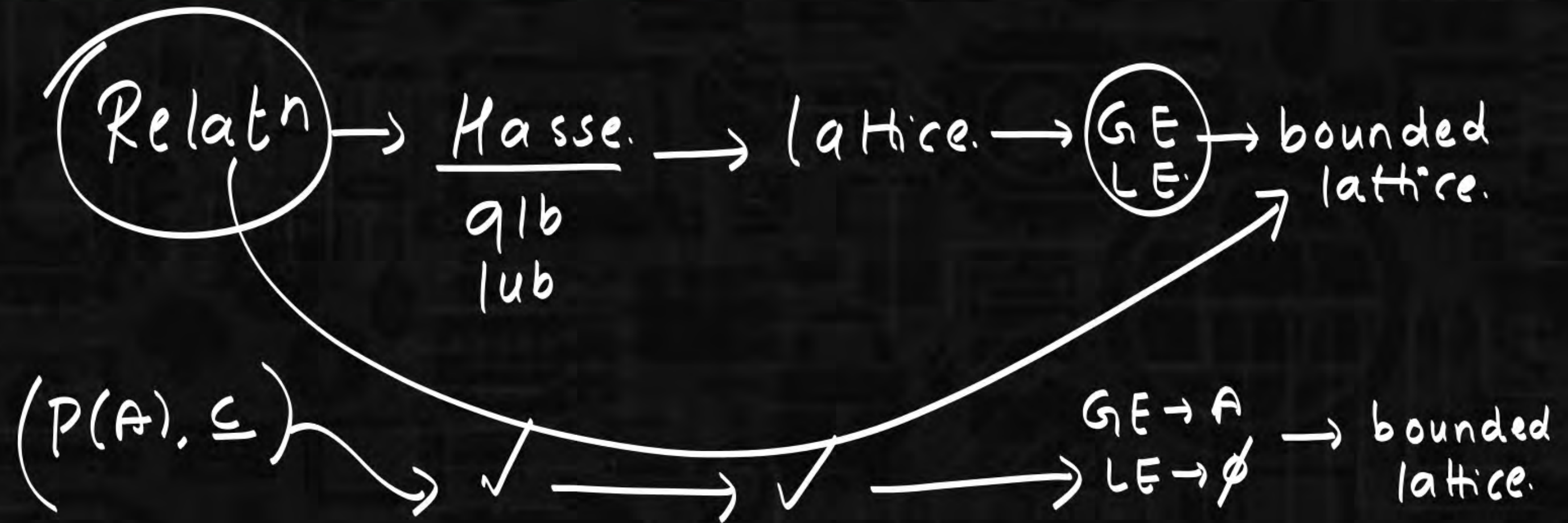
Greatest element $\rightarrow 1$ least element $\rightarrow 0$



if GE & LE are existing in lattice
then it is called bounded lattice.

eg: $(P(A), \subseteq)$ $GE \rightarrow A$ $LE \rightarrow \emptyset$ ✓
 $(\mathbb{Z}^+, \leq)$

Relations



Relations



$(\mathbb{Z}^+, \leq) \rightarrow \text{chain} \rightarrow \text{lattice} \rightarrow \begin{matrix} \text{GE} \rightarrow \text{NA} \\ \text{LE} \rightarrow 1 \end{matrix} \right)$ not bounded lattice.

$(\mathbb{Z}, \leq) \rightarrow \text{lattice} \rightarrow \begin{matrix} \text{GE} \rightarrow \text{NA} \\ \text{LE} \rightarrow \text{NA} \end{matrix}$

$\rightarrow$ if lattice is finite then it is bounded lattice.

g/b
l/b

Relations

BL $\rightarrow$ Complement lattice:

$$1 \rightarrow GE \quad 0 \rightarrow LE$$

$$a + b = 1 \quad lub(a, b) = GE$$

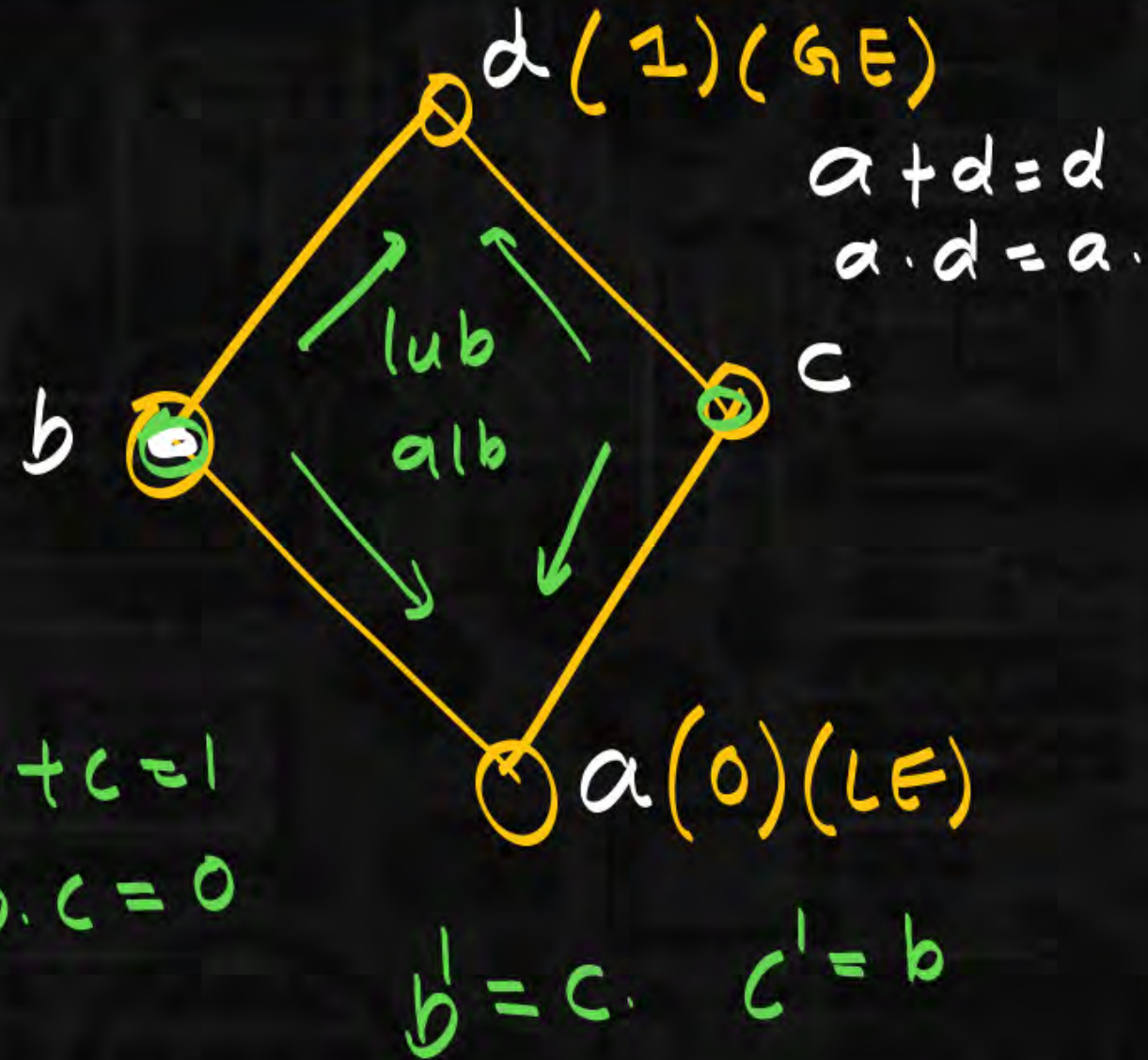
$$a \cdot b = 0 \quad glb(a, b) = LE$$

a & b are complement to each other.

$\rightarrow$ bounded lattice is complement lattice, when all elements will have at least 1 complement.

Relations

$$LE \rightarrow a(0) \quad GE \rightarrow (d)(1)$$



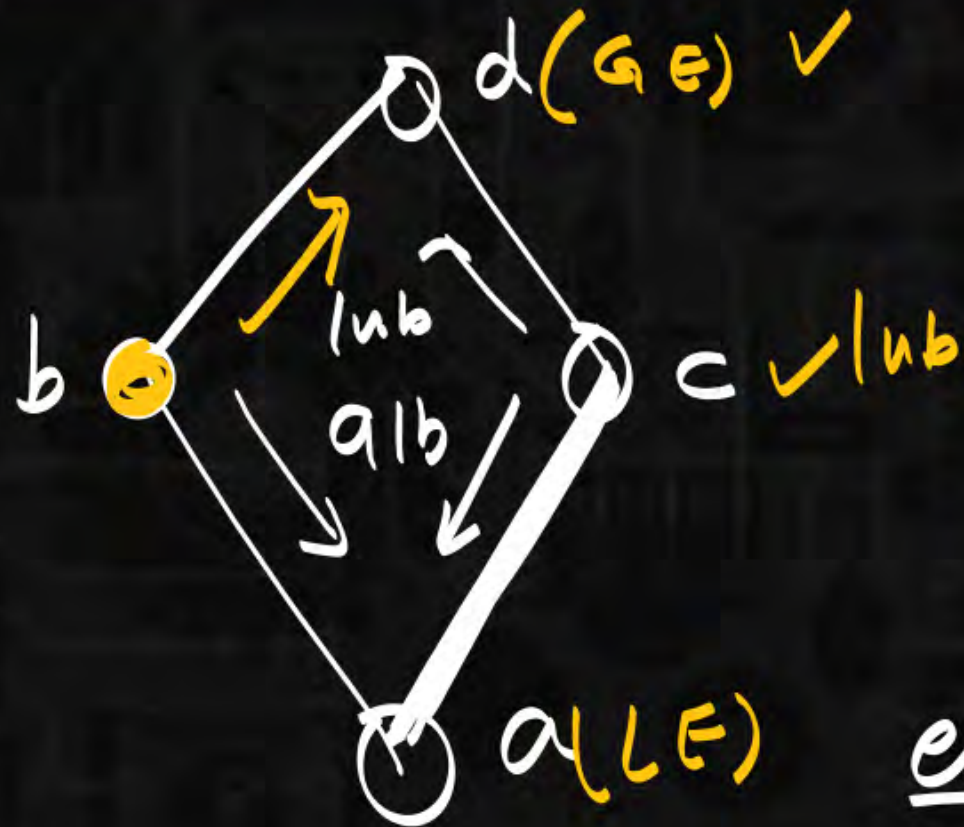
$$b + _ = 1(GE)(d)$$

$$b \cdot _ = 0(LE)(a)$$

$$\begin{aligned}
 b' &= c \\
 c' &= b \\
 a' &= d \\
 d' &= a
 \end{aligned}$$

$$\begin{aligned}
 b + _ &= d & b \cdot _ &= a \\
 \text{lub}(b, c) &= d & \text{glb}(b, c) &=
 \end{aligned}$$

Relations



$$b + _ = GE$$

$$b \cdot _ = LE$$

$$b + c = d$$

$$b \cdot c = a$$

$$\underline{lub(b, c) = d} \checkmark$$

$$\underline{glb(b, c) = a} \checkmark$$

ex: put c

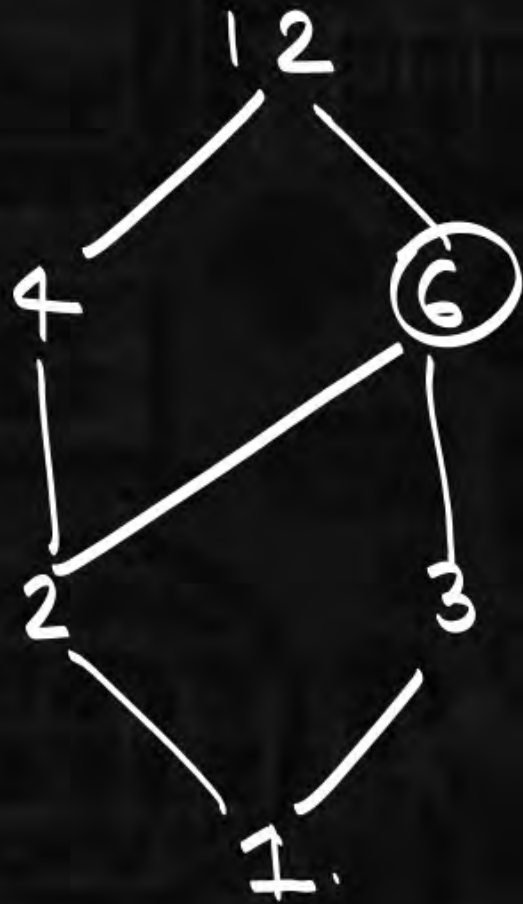
$$b' = c$$

Relations

$$GE \rightarrow 12 \quad LE \rightarrow 1.$$

$$(D_{12,1})$$

$$\left(\{1, 2, 3, 4, 6, 12\}, 1 \right)$$



for 4.

$$4 + = 12$$

$$4 \cdot = 1$$

$$\text{lub}(4,) = 12$$

$$\text{qlb}(4,) = 1.$$

$$\text{lcm}(4, 3) = 12$$

$$\text{gcd}(4, 3) = 1.$$

$$6 + = 12$$

$$6 \cdot = 1.$$

$$\text{lub}(6,) = 12$$

$$\text{qlb}(6,) = 1.$$

$$\text{lcm}(6,) = 12$$

$$\text{gcd}(6,) = 1.$$

④

$$4' = 3$$

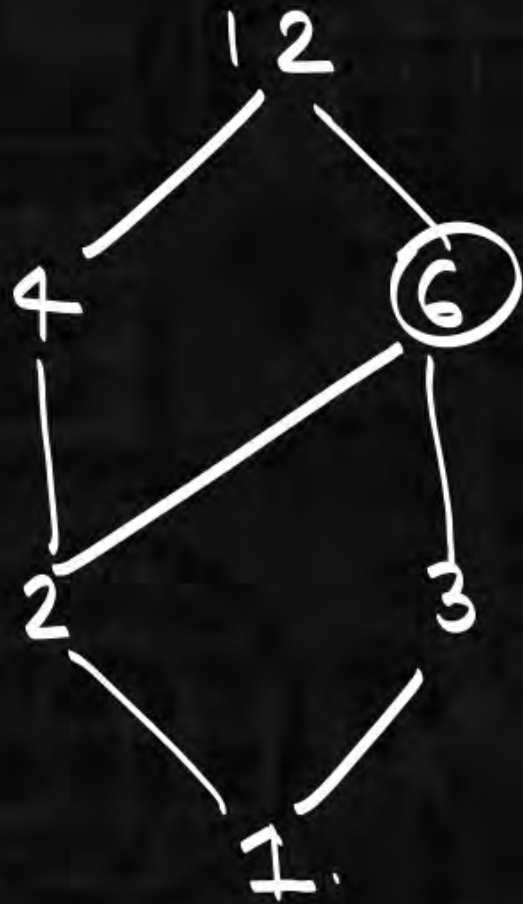
$$3' = 4.$$

Relations

bounded lattice ✓

$(D_{12}, |)$

$(\{1, 2, 3, 4, 6, 12\}, |)$



$$6 + = 12$$

$$6 \cdot = 1$$

$$\text{lub}(6,) = 12 \quad \text{glb}(6,) = 1$$

$$\text{lcm}(6,) = 12 \quad \text{gcd}(6, -) = 1$$

for 4

$$\text{lcm}(6, 4) = 12 \quad \text{gcd}(6, 4) = 1 \quad \times$$

for 3

$$\text{lcm}(6, 3) = 12 \quad \text{gcd}(6, 3) = 1 \quad \times$$

$$6' = \text{N.A.}$$

not complement lattice

Relations



$$b + c = GE \quad b \cdot c = LE$$

$$b + d = n$$

$$lub(b, d) = n$$

$$b' = d$$

$$b \cdot d$$

$$glb(b, d) = a$$

$$b' = \{c, d\}$$

$$c' = \{b, d\}$$

$$d' = \{b, c\}$$

$$a' = d$$

$$d' = a$$

Relations

$$A = \{1, 2, 3\}$$

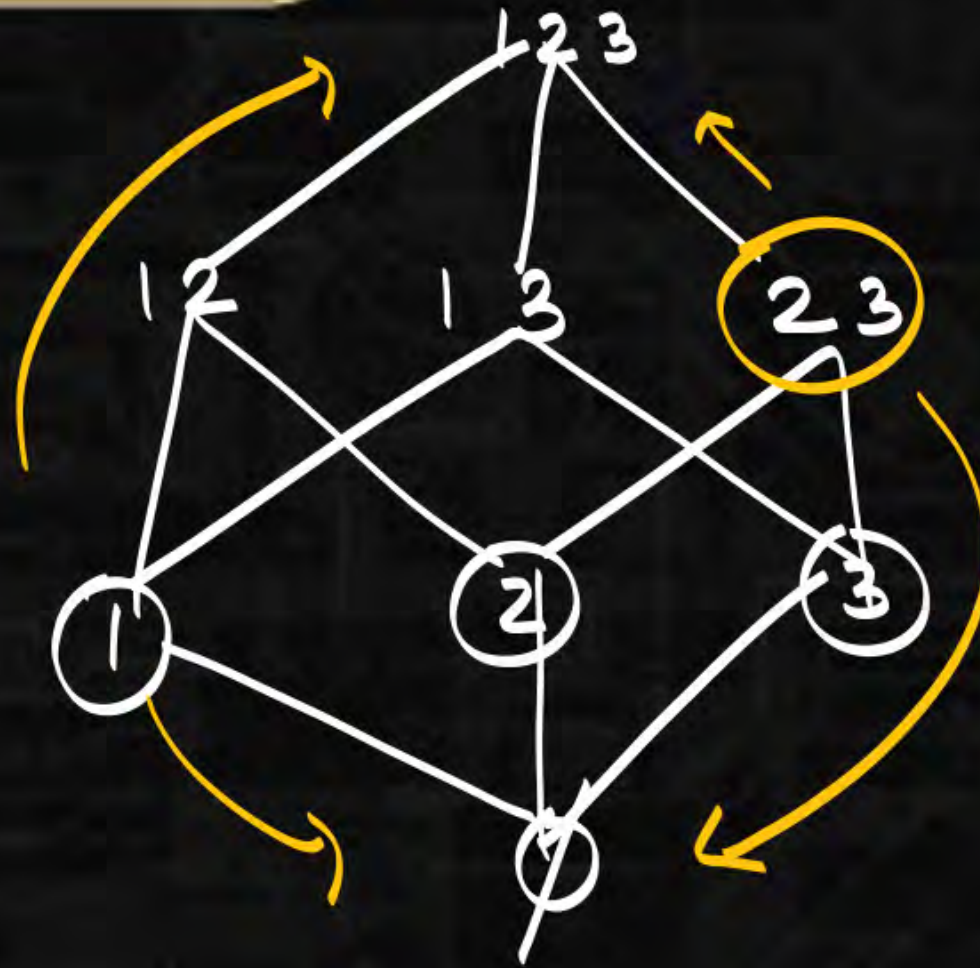
$$(P(A), \subseteq)$$

$$+ \rightarrow \cup$$

$$\circ \rightarrow \cap$$

$$1 + = 123$$

$$1 \circ = \emptyset$$



$$1' =$$

$$\{1\} \cup \{2,3\} = \{1,2,3\}$$

$$\{1\} \cap \{2,3\} = \emptyset$$

$$\{1\}' = \{2,3\}$$

Relations



Distributive lattice :

$$\text{Lattice} + \left\{ \begin{array}{l} a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) \\ a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) \end{array} \right.$$

Relations

(D12,1)



$$6 \wedge (4 \vee 2) = (6 \wedge 4) \vee (6 \wedge 2)$$

$$\frac{\text{lub}(4,2)}{\downarrow}$$

$$6 \wedge 4$$

$$\frac{\text{glb}(6,4)}{\downarrow}$$

$$2$$

$$\frac{\text{glb}(6,4)}{\downarrow} \vee \frac{\text{glb}(6,2)}{\downarrow}$$

$$\frac{2 \vee 2}{\downarrow}$$

$$= 2$$

234

$$6 \vee (4 \wedge 2) = (6 \vee 4) \wedge (6 \vee 2)$$

$$\frac{6 \vee 2}{\downarrow}$$

$$6$$

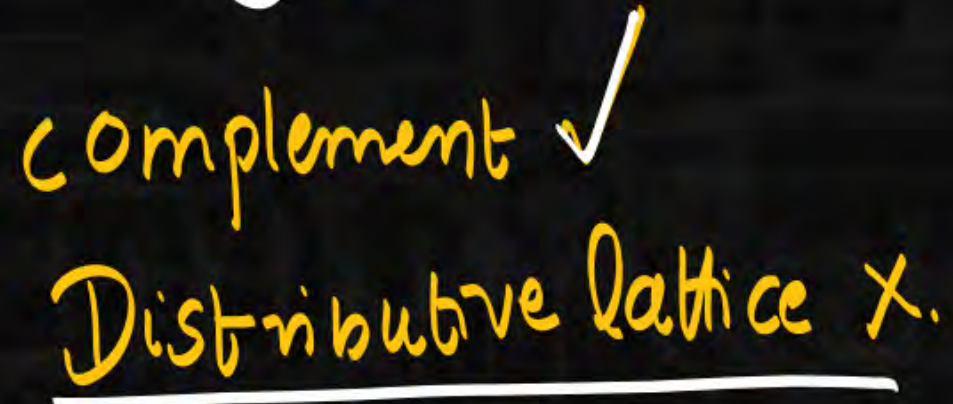
$$\frac{12 \wedge 6}{\downarrow}$$

$$= 6$$

Relations

$(D_{12}, 1) \rightarrow$ not complement lattice $6^1 = NA$
 $\rightarrow$ Distributive lattice. ✓

$(D_n, 1) \rightarrow$ distributive lattice.



$$x \vee (\underline{y \wedge z}) = (\underline{x \vee y}) \wedge (\underline{x \vee z})$$

$\text{glb}(y, z)$ $\text{lub}(x, y)$ $\wedge$ $\text{lub}(x, z)$
 $\downarrow$ $\downarrow$ $\downarrow$
 $x \vee a$ b $\wedge$ b
 $\text{lub}(x, a)$ $\underline{\hspace{2cm}}$
 $\downarrow$ b
 x

$$\neq$$

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$$

Relations

$+ \rightarrow \cup \rightarrow \vee \rightarrow \text{lub}$
 $0 \rightarrow \cap \rightarrow \wedge \rightarrow \text{glb}$



Boolean algebra = Lattice + complement + Distributive.

$$a + a = a$$

$$a \cdot a = a$$

$$a + b = b + a$$

$$a \cdot b = b \cdot a$$

$$a + (b + c) =$$

$$(a + b) + c$$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

$$a + (a \cdot b) = a$$

$$a \cdot (a + b) = a$$

$$a + b = 1$$

$$a \cdot b = 0$$

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

$$a + (b \cdot c) = (a + b) \cdot (a + c)$$

Relations

$$A = \{1, 2, 3\}$$

$$(P(A), \subseteq) \rightarrow \underline{\text{Boolean algebra}}$$

$$\begin{aligned} \bigcup &\rightarrow \text{lub} \\ \bigcap &\rightarrow \text{glb} \end{aligned}$$

Relations

$$\emptyset \rightarrow 000$$

